BSH/S-24/AI/DS/CS2T004/02

JAIDEV EDUCATION SOCIETY'S

# J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Education to Eternity

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

B.Tech / ~~M.Tech~~ / ~~MBA~~ / ~~MCA~~ Summer / ~~Winter~~ – 20[redacted]-24

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: CSE / DS / AI — Sem:- II

Subject: Basic Civil and Mechanical Engineering — Marks: 100

Subject Code: CS2T004 / DS2T004 / AI2T004

Date of Exam: 05/06/2024 — Time: 3 Hrs.

Instructions:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

COs: Students should be able to:

1. Define basic stream of Mechanical & Civil Engineering.
2. Explain the concepts of product manufacturing, Energy engineering, design engineering, Automobile engineering, construction technique and civil surveying.
3. Apply Basic knowledge of Casting, Machining, Designing, Manufacturing & Civil Construction technique.
4. Analyzed the different mechanical system and properties of construction & surveying material.
5. Interpret the problem in mechanical system and civil structure.

## PARTA

| Q. No. | Solve the two words or two sentences (All Questions are compulsory). | Marks | Bloom's Level and CO |
|---|---|---|---|
| Q.1 | Define Thermodynamics? | 2 | 1 |
| Q.2 | How engineering material is classified? | 2 | 1 |
| Q.3 | List any four different types of Power Plants? | 2 | 1 |
| Q.4 | What are the different types of drilling machines? | 2 | 2 |
| Q.5 | How the geological rocks can be classified? | 2 | 2 |
| Q.6 | Define Entropy term in thermodynamics. | 2 | 1 |
| Q.7 | What do you mean by reaming operation? | 2 | 1 |
| Q.8 | List any four civil engineering materials? | 2 | 1 |
| Q.9 | Which types of instruments are used for levelling in civil work? | 2 | 1 |
| Q.10 | List any four components of Automobile? | 2 | 1 |
| Q.11 | List any four operations performed on Lathe Machine? | 2 | 2 |
| Q.12 | List any four branches of mechanical engineering? | 2 | 2 |



Scanned with OKEN Scanner

## PART B

| Q.No. | Solve the following Descriptive Type Questions (All Questions are compulsory). | Marks | Bloom's Level and CO |
|---|---|---|---|
| Q.13 | Explain the difference between manufacturing and machining? | 4 | 1 |
| Q.14 | What are the main parts included sub structure of a bridge? | 4 | 1 |
| Q.15 | How are classified and identify column of a building structure? | 4 | 1 |
| Q.16 | Describe nuclear fission reaction. | 4 | 2 |
| Q.17 | What is the principal of machining with neat labelled diagram? | 4 | 2 |
| Q.18 | Explain the properties of civil engineering any four materials? | 4 | 1 |

## PART C

| Q. No. | Solve the following Descriptive Type Questions (Solve any 04 out of 06). | Marks | Bloom's Level and CO |
|---|---|---|---|
| Q.19 | Describe the main component of internal combustion engine with its neat labelled diagram. | 7 | 1 |
| Q.20 | Describe the production system with flow chart. | 7 | 1 |
| Q.21 | Explain the main parts of sensitive drilling machine with its neat labelled diagram. | 7 | 1 |
| Q.22 | Explain different steps involved in casting process in detail? | 7 | 2 |
| Q.23 | Explain the four Laws of Thermodynamics with appropriate examples? | 7 | 2 |
| Q.24 | How steam turbine power works and draw its neat labelled diagram? | 7 | 1 |

## PART D

| Q. No. | Solve the following Descriptive Type Questions (Solve any 02 out of 03). | Marks | Bloom's Level and CO |
|---|---|---|---|
| Q.25 | State any eight differences between Load bearing structure and framed structure. | 12 | 2 |
| Q.26 | Explain in details lathe machine's main parts with its neat labelled diagram. | 12 | 2 |
| Q.27 | State the different types of cement? | 12 | 1 |

acid with labour quick setting rapid hardening expanding

Scanned with OKEN Scanner